Paralysis Hand Recovery Machine – System Plan

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0) Quick Summary

A 3-phase rehab device (exoskeleton glove + forearm/wrist rig) that supports:

Phase 1 (Passive/Robot-driven): Safely moves the patient's hand in three modes - finger flex/extend (fist), wrist ulnar/radial deviation (left/right), wrist flex/extend (forward/backward).

Phase 2 (Active/Patient-driven): Measures the patient's voluntary capability in the same three DoFs without assistance (or with minimal assist-as-needed).

Phase 3 (Analytics & ML): Logs time, angle, torque/force, repetitions, and pain/effort ratings; visualizes progress with dynamic report generation using Power BI/Tableau or use Blynk; estimates recovery time using a simple and explainable ML model, upgraded later to more advanced models.

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1) Clinical & Safety Notes (non-negotiables)

- **Therapist-approved ROM limits:** Per-patient configurable limits for finger flex/extend, wrist flex/extend, and ulnar/radial deviation. Hard mechanical stops + software limits.
- **Torque/Force ceilings:** Current limiting on motors + real-time torque/force monitoring. Abort if exceeding threshold for >100 ms.
- **Emergency stop:** Big red e-stop on device + software e-stop in UI. Power stage de-energizes, brakes engage (if available).
- **Smoothness:** Jerk-limited trajectories (S-curves) to avoid sudden pulls.
- **Comfort & hygiene:** Soft liners, adjustable straps, easily cleanable surfaces.
- **Data privacy:** Local encryption at rest; export with patient consent.
- **Regulatory pathway (future):** Design toward IEC 60601-1 (electrical safety), 60601-1-2 (EMC), risk mgmt per ISO 14971, usability per IEC 62366, basic software lifecycle per IEC 62304.
- **Clinical validation:** Start with IRB/ethics approval and small feasibility study before claiming clinical efficacy.

1a) Initial Calibration & Setup (First Use Only)

- **Manual Calibration Routine:** On first setup, the therapist/caregiver manually moves the patient's hand to their maximum safe range in each DoF:
    - Fully open → fully closed fist (finger flex/extend).
    - Wrist forward → wrist backward (flex/extend).

    - Wrist left → right (ulnar/radial deviation).
- **Recording Limits:** The system records encoder/IMU angles at each extreme and saves them as per-patient software limits.
- **Safety Margins:** A 5–10° margin is automatically applied to avoid overstretching.
- **Persistent Storage:** Limits are stored in patient profile (JSON/EEPROM/DB) and automatically loaded in every session.
- **Re-calibration:** Can be repeated at any time if patient's ROM improves or straps/device are repositioned.

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2) System Architecture

Mechanical

- **Finger actuation** (1–5 DoFs depending on budget): tendon-driven cable system per finger or a single flexion/extension bar for all fingers (MCP-dominant).
- **Wrist module:**
    - Flexion/extension hinge with adjustable axis alignment to wrist joint.
    - Ulnar/radial deviation joint (orthogonal axis).
    - Quick-release mounts and soft straps.

Electronics

- **Controller:** STM32, ESP32-S3, or similar for low-latency loops; optional single-board computer (Raspberry Pi/Jetson) for UI + ML.
- **Drivers:** Current-controlled BLDC/servo drivers (e.g., ODrive/MadgeTech alternatives) or high-quality hobby servos with feedback for PoC.
- **Sensors:**
    - **Encoders** on each joint/tendon spool (≥12-bit).
    - **Force/torque:** In-line load cell or FSRs in glove fingertips + wrist strap; optional 6-axis force-torque for advanced control.
    - **IMU** on hand backplate to sanity-check angles.
    - **EMG** for bio-intent detection.

User Controls & Feedback:

- **Potentiometer (speed control):** Adjustable knob to scale exercise speed (rep cadence, trajectory velocity). Read via ADC on MCU.
- **LCD Display (status feedback):** Small LCD/OLED (e.g., 20×4 character LCD or 1.3" OLED) to show:
    - Current exercise/mode (Passive / Active / Assessment)
    - Repetition count & speed level

    - Error/safety warnings (e.g., torque exceeded, E-stop triggered)
    - Patient progress (ROM reached vs. target in real time)

Software

- **Low-level control:** 1 kHz inner loop (position/torque) on MCU; jerk-limited trajectory generator.
- **Mid-level modes:** Passive, Active-assist, and Assessment routines.
- **High-level UI & logging:** Desktop/web app (Python + FastAPI + Qt/Web) to set protocols, visualize, and export CSV/Parquet for BI tools.

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3) Degrees of Freedom (DoFs) & Modes

DoFs covered:

1. **Finger flexion/extension** (make/relax fist) — single coupled DoF initially.
2. **Wrist ulnar/radial deviation** (left/right).
3. **Wrist flexion/extension** (forward/backward).

Modes

- **Phase 1: Passive (robot-driven)**
    - Predefined ROM and cadence (e.g., 10 reps × 3 sets).
    - **Impedance control** (stiffness + damping) to remain compliant while tracking target trajectory.
- **Phase 2: Assessment (patient-driven)**
    - **No active torque** (or minimal gravity compensation).
    - Measures max angle, time-to-target, smoothness (jerk), and tremor power.
    - **Active-assist variant:** If patient stalls > T seconds below threshold force, provide minimal assist.
- **Phase 3: Analytics & ML**
    - Aggregates per-session metrics; estimates time-to-target recovery levels with survival regression or learning-curve models.

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4) Control Strategy (concise)

Trajectory generation:

- S-curve (jerk-limited) profiles between angle waypoints; per-patient ROM limits.

Impedance control (Phase 1):

- $\tau_cmd = K(\theta_ref - \theta_meas) + D(\omega_ref - \omega_meas) + \tau_ff$ (gravity/friction).
- Tune K/D conservatively; watchdog for high τ or error.

Assessment logic (Phase 2):

1. Cue target (angle or %ROM).
2. Start timer; record peak angle, time to reach 80% target, path variability, tremor index (RMS of 4–12 Hz band).
3. If **assist-as-needed**: add τ_assist only if $d\theta/dt < \varepsilon$ for $> T_stall$, capped at τ_max.

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5) Key Metrics to Log

- **Kinematics:** angles (°), angular velocity, jerk; ROM used (%).
- **Kinetics:** commanded torque/current (proxy), measured force/torque (N/N·m).
- **Performance:** time-to-target (s), success rate (% reps completed), repetitions, hold time.
- **Quality:** movement smoothness (spectral arc length), tremor power (RMS 4–12 Hz), path deviation.
- **Subjective:** pain (0–10), effort (Borg scale), fatigue flags.
- **Session metadata:** patient ID (pseudonymized), therapist ID, device config, firmware version.

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6) Data Schema (for CSV/DB)

patients

- patient_id, sex, age, side (L/R), diagnosis, baseline_date

sessions

- session_id, patient_id, date, therapist_id, mode (passive/assessment/assist), notes

trials (one per DoF/target in a session)

- trial_id, session_id, dof (fingers/wrist_fe/wrist_ud), target_angle_deg, target_type (abs/%ROM),
- start_time, end_time, success (bool), time_to_80pct_s, max_angle_deg, assist_applied (bool), pain_0_10

signals (optional large table)

- trial_id, t_ms, angle_deg, velocity_dps, torque_cmd_nm, force_n, imu_quat_wxyz

device_config

- config_id, fw_version, K_impedance, D_impedance, max_current, rom_limits

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7) ML Approach for Recovery Time Estimation

Goal: Predict **time to reach predefined functional milestone** (e.g., 80% baseline ROM, or therapist-set goal) using early-session data.

Baselines (start simple):

- **Linear regression / Elastic Net** on features from first N sessions → predict remaining sessions to goal.
- **Gradient Boosting (XGBoost/LightGBM)** for nonlinearities and interactions.
- **Survival analysis** if censoring exists (patient drops out): Cox proportional hazards or GBM-based survival models.

Example features (per patient up to session k):

- Trend of time-to-target (slope), best ROM so far, last-session smoothness, tremor index change, average assist torque, session adherence.
- Demographics & diagnosis (if available and approved).

Model hygiene:

- Train/validation split by **patient** (no leakage).
- Report MAE for "sessions remaining" and calibration.
- Keep **explainability** (feature importances, SHAP) for clinician trust.

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8) Visualization & Reporting

- **Live session dashboard:** per-rep angle/torque traces, safety flags, and rep counter.
- **Progress report:** weekly chart of ROM, time-to-target, smoothness, assist %.
- **Outcome estimate:** funnel chart with predicted recovery window and confidence band.

Power BI / Tableau data model:

- Import patients, sessions, trials (and optionally signals).
- Create measures: Avg Time-to-80%, Best ROM, %Sessions with Assist, Trend lines.
- Precompute tidy CSVs after each session for plug-and-play refresh.

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9) Bill of Materials (MVP options)

- **Mechanical:** 3D-printed wrist module + aluminum hinges; Bowden cables for finger actuation; soft glove.
- **Actuators:**
    - **Wrist FE/UD:** Small BLDC gimbal motors with planetary gearboxes (ratio 20–50:1) + magnetic encoders.
    - **Fingers:** High-torque servo (e.g., Dynamixel X-series) or BLDC with tendon spool.

- **Drivers:** Dual-channel BLDC/servo drivers with current sensing; emergency cutoff relay.
- **Sensors:** Inline load cell (0–50 N), FSRs for fingertip contact, 6-DoF IMU, 12–14-bit encoders.
- **Compute:** ESP32-S3 (control) + Raspberry Pi 4/5 (UI/ML) or single Jetson Nano/Orin Nano.
- **HMI:** 7–10" touchscreen or laptop; physical e-stop.

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10) Firmware/Software – Pseudocode Snippets

Impedance-controlled passive rep (Phase 1):

```
θ_target(t) = S_curve_profile(start=θ0, end=θ1, T=rep_time)
loop at 1 kHz:
  θ, ω = read_encoder()
  τ_limit = torque_ceiling(dof)
  τ_cmd = clamp(K*(θ_target-θ) + D*(ω_target-ω) + τ_ff, ±τ_limit)
  if force_sensor > F_max or |τ_cmd| near τ_limit for >100 ms: abort_to_safe()
  write_motor_current(τ_cmd)
  log(t, θ, ω, τ_cmd, force)
```

Assessment trial (Phase 2):

```
cue_user(target_angle)
start_timer()
while time < T_max and angle < 0.8*target:
  angle, vel = read_encoder()
  if assist_mode and stalled(vel, time):
    τ_cmd = min(τ_cmd + Δ, τ_assist_max)
  else:
    τ_cmd = 0
  log(t, angle, vel, τ_cmd)
end
metrics = {time_to_80pct, max_angle, tremor_power, smoothness}
```

Data pipeline (Phase 3):

```
after each session:
  compute per-trial metrics → write trials.csv
```

append sessions.csv summary

export tidy dataset for BI and for ML training

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11) Acceptance Tests (MVP)

1. **Safety cut-off:** Apply unexpected resistance; device must cut power < 100 ms.
2. **ROM accuracy:** Command +30° FE → measured within ±2°.
3. **Repeatability:** 100 passive reps: std dev of peak angle < 2°.
4. **Assessment reliability:** Time-to-80% repeatability CV < 10% in healthy subject.
5. **Data integrity:** No missing timestamps; CSV schema validates.